## 0.1 Monotone Functions

**Theorem 0.1.1.** Let $f : (a,b) \to \mathbb{R}$ be a monotonically increasing function, and let $x \in (a,b)$. Then, $f(x+)$ and $f(x-)$ exist. Specifically,

$$\sup_{a<t<x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x<t<b} f(t)$$

Moreover, if $a < x < y < b$, then $f(x+) \leq f(y-)$.

*Proof.* Since $f$ is monotonically increasing, for all $t \in (a,x)$, $f(t) \leq f(x)$. Therefore, the set $\{f(t) \mid a < t < x\}$ is bounded above by $f(x)$. This implies that $\sup_{a<t<x} f(t)$ exists and satisfies $\sup_{a<t<x} f(t) \leq f(x)$.
Given $\varepsilon > 0$, by the property of supremum, there exists $\delta > 0$ such that

$$\sup_{a<t<x} f(t) - \varepsilon < f(x-\delta) \leq \sup_{a<t<x} f(t).$$

Since $f$ is increasing, for $x - \delta \leq t < x$, we have $f(x-\delta) \leq f(t) \leq \sup_{a<t<x} f(t)$, which implies

$$\sup_{a<t<x} f(t) - f(t) < \varepsilon \quad \text{for all } x - \delta \leq t < x.$$

Therefore, for any sequence $\{x_n\}$ in $(a,x)$, $x_n \to x$ implies $f(x_n) \to \sup_{a<t<x} f(t)$, which shows

$$\sup_{a<t<x} f(t) = f(x-).$$

Meanwhile, the second assertion $f(x+) \leq f(y-)$ for $x < y$ can be proven by observing:

$$f(x+) = \inf_{x<t<b} f(t) = \inf_{x<t<y} f(t) \quad \text{and} \quad f(y-) = \sup_{a<t<y} f(t) = \sup_{x<t<y} f(t).$$

□

**Corollary 0.1.2.** The set of points of discontinuity of a monotone function is at most countable.

*Proof.* Let $f : (a,b) \to \mathbb{R}$ be a monotone increasing function, and let $E$ be the set of its points of discontinuity.
Since $f$ increases monotonically, $x \in E$ if and only if $f(x-) < f(x+)$.
By the density of rational numbers, for each $x \in E$, we can choose a rational $r(x) \in \mathbb{Q}$ such that

$$f(x-) < r(x) < f(x+).$$

If $x_1, x_2 \in E$ with $x_1 < x_2$, then from the previous theorem, we have:

$$r(x_1) < f(x_1+) \leq f(x_2-) < r(x_2).$$

This implies $r(x_1) < r(x_2)$, which shows that the map $r : E \to \mathbb{Q}$ is injective, thus $E$ is at most countable. □

**Corollary 0.1.3.** Monotone function has no Second-kind discontinuity.

## 0.2 Intermediate Value Property and Discontinuity

> **Theorem 0.2.1.** If $f : (a, b) \to \mathbb{R}$ satisfies the Intermediate Value Property (IVP), then $f$ cannot have a discontinuity of the First-kind discontinuity.

*Proof.* Suppose that $f$ has a First-kind discontinuity at $x \in (a, b)$.
That is, by definition, $f(x^-)$ and $f(x^+)$ both exist, but the equality $f(x^-) = f(x) = f(x^+)$ does not hold.
WLOG, suppose $f(x^-) \neq f(x)$. Let $\varepsilon = |f(x^-) - f(x)| > 0$. Since $f(x^-)$ exists, there exists $\delta > 0$ such that

$$t \in (x - \delta, x) \implies |f(x^-) - f(t)| < \frac{\varepsilon}{3}.$$

Now, consider the case where $f(x) > f(x^-)$. For any $t \in (x - \delta, x)$, we have:

$$f(t) < f(x^-) + \frac{\varepsilon}{3} = (f(x) - \varepsilon) + \frac{\varepsilon}{3} = f(x) - \frac{2\varepsilon}{3}.$$

In particular, for $t_0 = x - \frac{\delta}{2}$, we have $f(t_0) < f(t_0) + \frac{2\varepsilon}{3} < f(x)$. By the Intermediate Value Property, for the value $y = f(t_0) + \frac{2\varepsilon}{3}$, there must exist some $c$ between $t_0$ and $x$ such that $f(c) = y$.
However, for any $c \in (t_0, x) \subset (x - \delta, x)$, we have:

$$|f(c) - f(t_0)| \leq |f(c) - f(x^-)| + |f(x^-) - f(t_0)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \frac{2\varepsilon}{3}.$$

This implies $f(c) < f(t_0) + \frac{2\varepsilon}{3} = y$, which contradicts $f(c) = y$.
If $f(x) < f(x^-)$, a similar contradiction arises. Therefore, $f$ cannot have a discontinuity of the first kind. $\square$